Unit 10 - Communication Technologies
Assignment 8: Upgrade a LAN Task 10

Introduction

Task description - *You are a junior consultant for T5 Alliance. Your new client, Competition Bikes Ltd, have decided that they would like to have wireless capability in their new network. You must adjust your design to include the following new requirements;*

- *Wireless coverage must extend to the whole site.*
- *The wireless network must be as secure as possible.*

Task 10

(U10D1) - Compare the OSI seven layer model and the TCP/IP model

Task description - *Comparison of the OSI and TCP layered models to include;*

- *Short description of each layer*
- *How the layers from each model map to each other*
- *The purpose of each of the two models*

Introduction

This section will aim to offer a degree of comparison between both the OSI 7-layer model, and the TCP/IP model. Which I will be stating and comparing the definition, purpose, similarities and differences of both the OSI and TCP/IP models.

In order to compare the two, a brief analysis of both their similarities and differences must be performed, as well as an explanation of their development and operation. With illustrative help from images and tables for easier explanation.

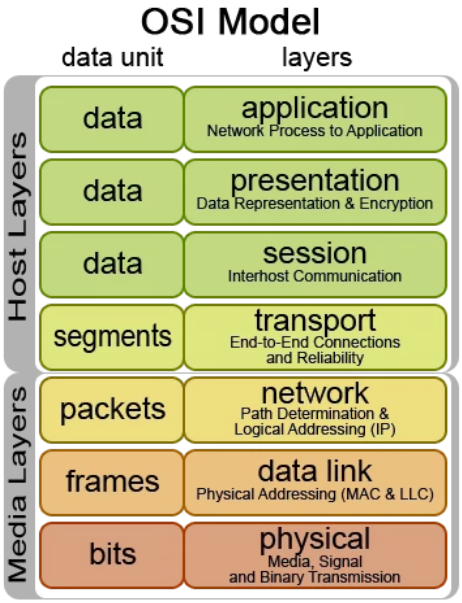
Short description of each layer

OSI Model

OSI stands for open systems interconnection model and is used by developers for discussing about how messages should be transmitted between certain points in any telecommunication network.

OSI was created by the International Organisation for Standardisation (ISO) to describe, analyse and work with multiple communication protocols used within the networking aspect of computing.

Additionally, it should be noted that OSI is regarded as being a “stricter” model that clearly defines the different layers required for network communication to take place, expanding upon the four layers of TCP/IP.



As seen in the image above, there are 7 different layers found in the OSI model with each one having a different functionality purpose. Basically, it contains layers so it makes the information transferred within a network much easier for devices within said network.

Below is a brief description of each of the layers, with the name of said layer and their function within the network.

Layer 1 - Layer 1 is the physical layer. This layer involves movement of data (Which are a collection of binary numbers in “bits”) through a network, allowing devices to send and receive data over a carrier network through cables, connections and network interface cards.

Layer 2 – Layer 2 is known as the data-link layer. The role of this layer is to create links across a network, which is done by gathering information and transferring data in the forms of frames. For instance, a header would contain the source of the information and a destination MAC address to know where the information is going.

This type of communication needs two sublayers in order for it to work efficiently, the first layer being: The Logical Link Control, which is used for the flow control of information and detecting a variety of possible errors.

With the second sublayer being the Media Access Control, or MAC, that is used for finding the hardware which is the destination of the data using a 12 byte hexadecimal number and controlling the data stored within the hard drive of the device.

Layer 3 – Layer 3 is known as the network layer. It is used for creating the address, the route of the data and determining the path which the data will travel through, ensuring that the information gets to the right place.

For example, this layer uses the IP addresses found in the device, which is the network layer for the internet, for identifying the host within a network.

Layer 4 – Layer 4 is known as the transport layer. It is used for managing the packages of data and the delivery of these packets to the correct location. For instance, it does this by using flow control and data recovery, which could occur in the device before and after transmission to make sure the data has been delivered properly.

Layer 5 – This layer is known as the session layer. The role of this layer is to handle conversations, and this involves the start and ending of conversations. This includes the devices connecting and the reconnection once after an interruption.

Layer 6 – Layer 6 is known as the presentation layer, which should be a part of an operating system, such as Windows, Linux, etc. The role of this layer is for data compression and encryption, which means it converts incoming and outgoing data on the network into a smaller data size when sending and readable presentation format when received.

Layer 7 – This layer is known as the application layer and it is the final layer in the model. The role of this layer is to allow network applications to be able to communicate with other network applications. Basically, it identifies partners within a network and therefore allows data send to be opened and shown on other partners.

TCP/IP Model

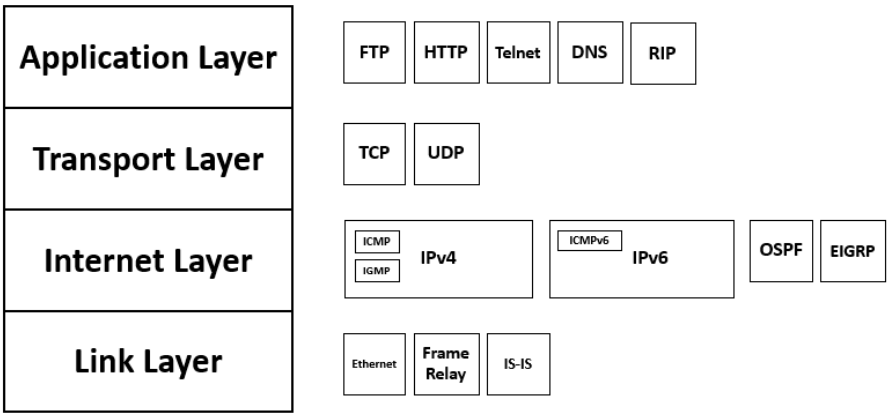
The TCP/IP (Transmission Control Protocol/Internet Protocols) model was developed by DARPA (Defense Advanced Research Project Agency) during the 1970s. Which is part the US Dept. of Defence, and is maintained by the IETF (Internet Engineering Task Force).

In simple terms, the TCP/IP model contains a complete suite of protocols which are used on the internet to connect hosts on the World Wide Web. To elaborate, the TCP element of the model defines the actual transmission of the data between applications on a network, with the IP element defining how data is routed and managed through the network.

In the modern day, TCP/IP is built into most operating systems, making it the standard for the transmission of data over networks. There are certain cases that network operating systems also have their own protocols, an example of this would be NetWare, however operating systems similar to this can also support TCP/IP.

However, TCP/IP does not offer basic support for protocols such as FTP, HTTP, with these having to be built on top of the pre-existing TCP/IP model. This is due to the model being developed a long time ago, and for a specific purpose, whereas OSI is intended to be a much more generalised model.

Basically, this model provides end to end connectivity which shows how data over a network should be packeted, addressed, transmitted and routed in order for the data to be controlled and received at its final destination.



As seen in the picture above, the TCP/IP model contains layers just like the OSI model. It exists to define the general guidelines for the design and implementation of protocols found in computers connected to a network, also like the OSI model. However, unlike the OSI model, it only contains 4 layers instead of 7 layers, but this does not mean that they don't correlate to each other well.

Below are brief descriptions of the four layers found in the TCP/IP model, with their name and purpose listed with some examples.

Layer 1 - The first layer, or the bottom one, of the model is the Network Access/Interface Layer. This is the layer that makes sure that the physical components of a network are connected to both the network and the Internet Protocol. Basically, it deals with all the hardware in the network between the network and the IP.

Layer 2 - The second layer, or the second to bottom one, is the Network/Internet Layer. The purpose of this is layer is the pack data into IP datagrams which allows them to forward between and across networks. Essentially, it manages the transmission of data between two devices found in a network.

Layer 3 - The third layer, or the second to top one, of this model is the Transport Layer. The purpose of the layer is to allow devices to control and maintain a conversation through the transportation of data. Basically, it manages the consistency of the data traffic between two hosts so that the data arrives at the destination host.

Layer 4 - The last layer, or the top one, of the model is the Application Layer. This is the layer that defines the details of how data is sent through the network. Essentially, this layer defines TCP/IP protocols and how programs are hosted within a network, an example of this would be the Domain Naming System (DNS).

How the layers from each model map to each other / The purpose of each of the two models

As previously discovered, the 4-layer structure of TCP/IP is built as information is passed down from applications to the physical network layer. So, when data is sent, each layer treats all of the information it receives from the upper layer as data, adds control information, called a header, to the front of that data and then passes it to the lower layer. This essentially means when the data is received, the opposite procedure will take place as each layer processes and removes its header before passing the data to the upper layer.

With the OSI model, the 7 layer system can be considered the basis of network protocol structure, but it is not really good at being changed to be adapted to more modern standards.

So, later interpretations of protocols have been modified. Such as the TCP/IP layer models have combined, or re-structured, where protocols fit and therefore how data is moved between networks. The obvious difference being the merging of some of the OSI layers in order to simplify the whole protocol model.

For simplicity purposes, below is a comparison table between the two models, with each containing a much descriptive definition compared to the previous brief description. So that the combination of the OSI and TCP/IP model are seen at much better detail.

OSI layer	The TCP/IP 4-layer model	OSI - 7 Layers Model
Layer 7	**Application Layer (Layer 4)** The Application Layer in TCP/IP groups the functions of OSI Application, Presentation Layer and Session Layer. Therefore any process above the transport layer is called an Application in the TCP/IP architecture. In TCP/IP socket and port are used to describe the path over which applications communicate.	**Application Layer** · Defines interface-to-user processes for communication and data transfer in network · Provides standardized services such as virtual terminal, file and job transfer and operations
Layer 6		**Presentation Layer** · Masks the differences of data formats between dissimilar systems · Specifies architecture-independent data transfer format · Encodes and decodes data; encrypts and decrypts data; compresses and decompresses data
Layer 5		**Session Layer** · Manages user sessions and dialogues · Controls establishment and termination of logic links between users · Reports upper layer errors

Layer 4	**Transport Layer (Layer 3)** In TCP/IP architecture, there are two Transport Layer protocols. The Transmission Control Protocol (TCP) guarantees information transmission. The User Datagram Protocol (UDP) transports datagrams without end-to-end reliability checking.	**Transport Layer** · Manages end-to-end message delivery in network · Provides reliable and sequential packet delivery through error recovery and flow control mechanisms · Provides connectionless oriented packet delivery
Layer 3	**Network/Internet Layer (Layer 2)** (IP) is the primary protocol in the TCP/IP Network Layer. All upper and lower layer communications must travel through IP as they are passed through the TCP/IP protocol stack. There are many supporting protocols in the Network Layer, such as ICMP, to facilitate and manage the routing process.	**Network Layer** · Determines how data are transferred between network devices · Routes packets according to unique network device addresses · Provides flow and congestion control to prevent network resource depletion
Layer 2		**Data Link Layer** · Defines procedures for operating the communication links · Frames packets · Detects and corrects packets transmit errors
Layer 1	**Network Interface/Access Layer (Layer 1)** In the TCP/IP architecture, the Data Link Layer and Physical Layer are normally grouped together to become the Network Access layer. TCP/IP makes use of existing Data Link and Physical Layer standards rather than defining its own.	**Physical Layer** · Defines physical means of sending data over network devices · Interfaces between network medium and devices · Defines optical, electrical and mechanical characteristics

Conclusion

As seen in the table above, the four layers of the TCP/IP model have layers equivalent that to the OSI model, with the top three layers of the OSI model correlated with a single layer to the TCP/IP model.

However, they still have differences in what specific purposes their layers contain in order to fit for different scenarios, but not too different to not be dissimilar. Basically, both of the models above have similarities and differences which make them unique and better for certain purposes but can still be used together.

One of the similarities is that they are similar in the way they are structured. This is because both of the models are constructed with the use of layers and both contain layers which correlate with each other well.

Especially with the Application Layer, where it is roughly the same definition for both models. However, each model uses different services depending on which model is used. Both models

also have similar transport and network layers, this means that functions performed between these layers on each of the models will have a similar effect and outcome.

Another similarity of the two models with scenarios is that they both assume that the packets of data will be switched, meaning that individual packets of information which are sent over the network may take slightly different paths so they can reach the same destination.

One of the main differences between the two models is the amount of layers each have, the OSI model consists of 7 layers, whilst the TCP/IP only consists of 4 layers.

This is because the TCP/IP model combines both the presentation and the session layers, which can also be seen on the OSI model, into the application layer. Therefore making it shorter,
because of this the TCP/IP model seems the less complex of the models due to the fact that it has three fewer layers.

One of the major differences between the two models is that TCP/IP is the more credible of the two models, this is because this models and its protocols are the standard to which the World Wide Web was built, which therefore gives it credibility, this is a contrast to the OSI model, this is because this model only exists as a guidance tool to which models can be built around.

Overall, Both models were created independently from each other, hence why they have a few differences in purpose and amount of layers, but they both still have some similarities. For instance, the data link and physical layers on the OSI model, which are both added into the Network Interface/Access Layer of the TCP/IP model.

Also, there are similarities with the purpose for both models. Which is that there is a need to simplify the data flow process because there are so many different networks in use.

For instance, with the Internet, this flexibility in protocol structure is needed in order for these many networks to be able to share data. The reason being that accurate addressing of packets within a network is important, because it defines how the data is transmitted (or transported).

TCP/IP is also more progressive when compared to the OSI model. The reason for this is because the OSI 7 layer model is much less rigid, meaning that TCP/IP can change the interpretations of the protocol structures.

Hence why there are currently two TCP/IP models. Those two being a 4 layer model, which has been mentioned, and a 5 layer model, which hasn’t yet been mentioned. With the 5 step model adding an additional layer called the “Data Link Layer” for MAC addressing. Meaning that this is evidence that the protocol structure is a developing concept.